A Review of LLM Agent Applications in Finance and Banking

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Abstract

The accelerating digital transformation in finance and banking, coupled with advances in large language models (LLMs), has spurred investigation into LLM-powered computational agents capable of simulating, analysing, assisting, and acting within complex financial ecosystems. Leveraging their advanced reasoning and linguistic capabilities, these agents are uniquely positioned to address the multifaceted challenges of modern banking, finance, and economics, delivering scalable solutions for risk management, regulatory compliance, and strategic decision-making. Yet, the rapidly growing literature on LLM agents in the sector remains fragmented, lacking a cohesive survey evaluating their capabilities, risks, and real-world applicability. This survey paper offers a review of the current literature on LLM agents in finance and banking, categorising their applications in the sector into four core functions: simulation, acting, analysis, and advising. We examine a large corpus of studies employing LLM agents in market simulation, macroeconomic and microeconomic scenario planning, synthetic data generation, automated trading, and decision support systems among other applications, while critically analysing their technical efficacy, ethical dimensions, regulatory compliance, and operational limitations. In our survey we find that while LLM agents excel in linguistic tasks, their deployment in mission-critical systems requires hybrid architectures (including human supervision) and robust safeguards against hallucinations and biases. By integrating insights from diverse subject areas and frameworks, our work highlights key challenges and opportunities for advancing the safe and effective use of LLM agents in finance and banking. This work serves as both a reference for researchers and a pragmatic guide for practitioners navigating the transformative potential and pitfalls of LLM agents in finance and banking.

1 Introduction

The accelerating digitisation of banking operations, financial markets, and economic systems has fuelled a surge in efforts to automate routine tasks and augment human decision-making through advanced computational tools. Financial institutions now prioritise automating labour-intensive processes such as transactional reporting, compliance monitoring, and credit risk assessment to reduce operational latency and human error, while freeing resources for strategic analysis and decision-making [1, 2]. Simultaneously, augmentation technologies are being deployed to enhance predictive analytics, algorithmic trading, and personalised financial advising, where human expertise is amplified by real-time data processing and pattern recognition [3, 4]. Parallel to this shift, simulation continues to serve as a cornerstone for modelling complex, interconnected systems [5] from stress-testing portfolios under geopolitical shocks [6, 7] to simulating the

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effects of credit card promotions under diverse market scenarios [8] or simulating emergent behaviours in cryptocurrency markets [9]. Together, the dual focus on automation-augmentation and high-stakes simulation reflects a broader recognition: as banking operations become more targeted and economic environments grow more data-intensive, scalable computational frameworks are indispensable for resilience, innovation, and evidence-based policy-making.

Building on these foundations of automation, augmentation, and simulation, LLM-powered agents, i.e., AI systems built on large language models that can process language, reason, and act autonomously, represent a transformative leap in addressing the unresolved challenges of modern finance and banking. Traditional rule-based and classical machine learning systems, while effective on structured tasks, struggle to interpret unstructured data such as earnings call transcripts, regulatory filings, or geopolitical news that increasingly drives market volatility [10]. LLM agents have the potential to bridge this gap by combining natural language understanding with dynamic reasoning, enabling autonomous systems to parse ambiguous information, infer context, and generate actionable insights in real time. For instance, they can automate complex workflows like understanding customer complaints, looking up documentation relevant to the complaint, and generating a response to resolve the complaint. In trading, LLM-driven agents can enhance decision autonomy by integrating real-time sentiment analysis with probabilistic risk frameworks [11], while in retail banking, they can personalise customer interactions by dynamically adapting to evolving financial goals [12]. Crucially, their ability to simulate human-like interactions in agent-based models [13] could allow institutions to stress-test strategies against synthetic yet plausible market participants, from retail customers to algorithmic traders. By merging linguistic intelligence with computational rigour, LLM agents could not only operationalize autonomy in unstructured environments but also advance simulation fidelity.

While LLM-powered agents hold transformative potential for banking and finance, their adoption in mainstream systems hinges on rigorous, systematic, and scientific assessments of their accuracy, robustness, fairness, interpretability, and compliance with regulatory and risk management standards across applications [14–16]. A large yet fragmented body of work explores these possibilities investigating the use of LLM agents as financial advisories, experimental frameworks for multi-agent trading environments, and theoretical models of LLM-driven macroeconomic forecasting. However, these efforts are siloed and there is a lack of a review that consolidates these findings, making it challenging for researchers and practitioners to obtain a holistic understanding of the scope and readiness of LLM agents in financial applications. Furthermore, early experiments often prioritise technical feasibility over critical questions of limitations and challenges of applying LLM agents in the sector such as ethics and alignment with existing frameworks. This review addresses these gaps by analysing existing research, mapping emerging methodologies, and highlighting key challenges and opportunities for advancing the safe and effective use of LLM agents in finance and banking, separating computational novelty from tangible impact and real-world applicability.

To contextualise this review, we define an *agent* as a computational entity that perceives inputs, processes information, and takes actions to achieve objectives within an environment. In finance, agents have historically ranged from simple algorithmic traders executing predefined rules to adaptive systems that dynamically adjust strategies based on real-time data (e.g., portfolio rebalancing bots) [17]. Following on from this, *LLM agents* are AI systems that extend the agentic paradigm by integrating the linguistic and reasoning capabilities of LLMs, enabling them to interpret unstructured text, generate context-aware decisions, and act autonomously or semi-autonomously while interacting with other tools or databases, other agents or independently. In banking and finance, the applications of LLM agents can be structured along any of the following three categories:

1. **Agent function**: a) *Simulating*, i.e., whether the agent simulates scenarios, (e.g., simulating market behaviour using LLM-driven participants), b) *Analysing*, (e.g., detecting fraud patterns), c) *Advising*, i.e., provides recommendations, (e.g., personalised wealth advisor), or d) *Acting*, i.e., executes tasks (e.g.,

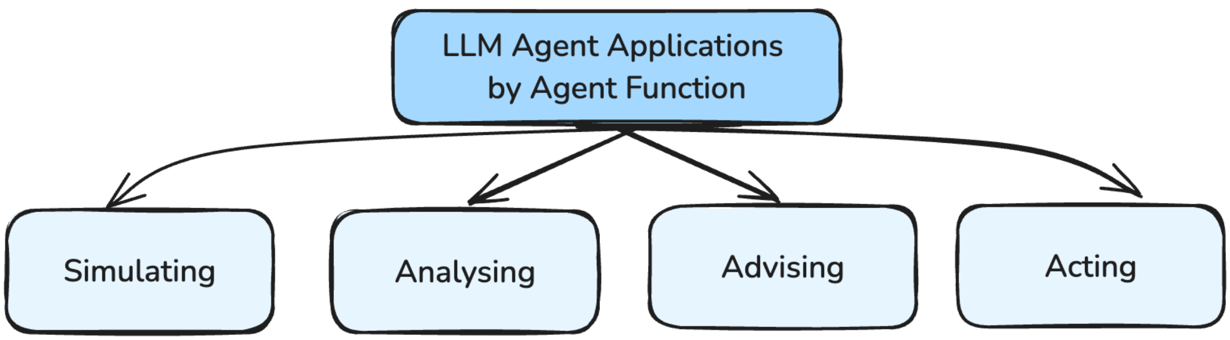

Figure 1: LLM agent applications by agent function

automated trading);

2. **Operational domain**: domains spanning retail and commercial banking (e.g., loan approval workflows), investment banking (e.g., mergers and acquisitions due diligence), and economics (e.g., macroeconomic forecasting);

3. **Architectural complexity (single-agent vs. multi-agent)**: where single agents operate in isolation (e.g., a retail chatbot), while multi-agent frameworks simulate interactions between heterogeneous actors, making decisions in cooperation (e.g., different agents owning different actions in a workflow) or competition (e.g., competing algorithmic traders) with other agents.

For the purpose of this paper, we have made the choice of structuring the reviewed works by agent function primarily because these categories directly map to the core challenges LLM agents are poised to address in finance: scenario testing (simulation), automating workflows (acting), enhancing analytical precision (analysis), and scaling human expertise (advising). This setup still allows for literature coverage across subject domains (e.g., retail banking vs. investment) and architectural frameworks (single- vs. multi-agent), while enabling systematic assessment of LLM agent readiness for real-world deployment, from technical feasibility (e.g., can an agent reliably act on unstructured data?) to economic value (e.g., does simulating adaptive agents improve risk forecasts?). While we cover a massive body of literature in the applications of LLM agents on finance, banking, and economics, any literature on applications in the sub-field of 'Insurance' are out of scope of this paper.

In this paper, we:

- consolidate an otherwise fragmented body of literature on the applications of LLM agents across finance, economics, and banking, offering a central reference point for both researchers and practitioners.
- introduce a clear four-part classification of LLM agent applications by their primary function: simulation, acting, analysis, and advising. This taxonomy serves as a structured lens for examining how these agents address core financial challenges, from scenario testing (simulation) to automation (acting), analytical precision (analysis), and decision support (advising), without discounting multiple subject domains (retail banking, investment banking, macroeconomic forecasting, etc.) and architectural frameworks (single- vs. multi-agent).
- highlight key technical, ethical, and regulatory constraints such as model interpretability, robustness, and compliance requirements that must be addressed to ensure LLM agents can be integrated responsibly and effectively into mission-critical financial systems.
- build on identified gaps, propose directions for advancing LLM agent robustness, interpretability, and ethical governance, including the development of standardised evaluation frameworks and adaptive policies

aligned with evolving financial technologies and regulatory landscapes.

The remainder of this paper is structured as follows. Section 2 provides a historical overview of AI applications in finance and banking, introduces foundational concepts and definitions relevant to AI and agent-based systems, and surveys the technical evolution and institutional reception of LLMs within financial contexts. Section 3 presents a systematic review of current research employing LLM agents in finance and banking, organised by functional taxonomy (simulation, acting, analysis, and advising). This section synthesises key findings, methodological innovations, and implications for both academic research and industry practice. Section 4 critically examines the limitations and challenges inherent to LLM agent deployment, including technical constraints, ethical challenges, and regulatory considerations, while evaluating their readiness for integration into mission-critical financial systems. Finally, Section 5 outlines future research directions, proposing frameworks for advancing LLM agent robustness, interpretability, and ethical governance, as well as novel applications at the intersection of generative AI and computational economics.

2 Background

2.1 AI in Finance and Banking

2.1.1 Early AI Applications in Finance and Banking

Early analytical applications in finance and banking were largely dominated by rule-based expert systems. For instance, FICO's rule-based scoring systems, introduced in the 1980s, became an industry standard by codifying expert heuristics into deterministic decision trees [18]. By the early 2000s, classical machine learning techniques focusing on structured tasks such as credit scoring, fraud detection, and basic predictive analytics gained further traction. Logistic regression models routinely started getting used to predict loan defaults based on historical repayment data [19], while support vector machines (SVMs) excelled in detecting fraudulent transactions by capturing non-linear patterns in payment histories [20]. Time-series forecasting models, such as ARIMA (AutoRegressive Integrated Moving Average), became popular for predicting stock prices, exchange rates, and macroeconomic indicators [21], while decision trees [19] and ensemble methods like random forests [22] improved credit risk assessment by modelling hierarchical splits in borrower attributes. Survival analysis techniques, including Cox proportional hazards models [23], were adapted to estimate customer churn probabilities and lifetime value in retail banking [24, 25]. Although these methods were well-suited to structured data such as balance sheets, transaction logs, and credit histories, they relied heavily on manually engineered features and lacked the flexibility to adapt to evolving market complexities and unstructured information streams.

2.1.2 Deep Learning in Finance and Banking

As financial institutions began digitising more processes from transactional record-keeping to customer complaint recordings, vast volumes of heterogeneous data became more readily available, and advances in cloud computing and graphics processing units (GPUs) provided the computational resources required to train significantly more complex models [26]. This confluence of abundant data and high-powered computing led to the adoption of deep learning algorithms, allowing networks to learn latent representations directly from raw inputs and enabling institutions to harness these advanced models for tasks requiring the processing of unstructured, high-dimensional data.

Building on these developments, Recurrent Neural Networks (RNNs) and Long Short-Term Memory networks (LSTMs) began being used for advanced time-series forecasting, capturing temporal dependencies in transaction sequences [27] and outperforming traditional ARIMA models in predicting stock volatility [28]. LSTMs also became a popular architecture for NLP tasks [29] before transformer-based LLMs took the centre stage. Convolutional Neural Networks (CNNs), given their image processing capabilities, started being actively used for counterfeit currency detection [30], biometric authentication and identity verification [31], cheque and handwritten document processing [32], and text and image-based document classification [33].

Other architectures like Deep Belief Networks (DBNs), which identified latent borrower risk factors overlooked by logistic regression, enhanced credit risk modelling [34] and market prediction [35], while autoencoders improved fraud detection [36, 37]. Generative adversarial networks (GANs) further addressed data scarcity by synthesising realistic financial scenarios for stress testing [38]. However, these models introduced challenges, including computational costs for GPU-intensive training and "black-box" decision-making that clashed with regulatory demands for transparency [39]. Despite these hurdles, deep learning laid the groundwork for adaptive, end-to-end systems capable of handling the scale and complexity of modern financial ecosystems setting the stage for more advanced and complex versions of paradigms like deep reinforcement learning and agent-based modelling.

2.1.3 Advanced AI in Finance and Banking

While deep learning-enabled systems extract latent patterns from unstructured data and model non-linear relationships at scale, these advancements have struggled to address the dynamic, interactive, and adaptive demands of modern financial ecosystems, where decisions unfold in real time amid shifting market conditions and multi-agent interactions. The desire to tackle these challenging problems spurred the integration of advanced paradigms such as reinforcement learning (RL), multi-agent systems (MAS), and agent-based modelling (ABM) into AI systems. These methods address issues with passive prediction to enable complex simulation and robust scenario and decision-making analysis. These methods are relevant to this paper and warrant a brief introduction.

Reinforcement Learning: RL paradigms enable systems to learn optimal decision-making policies through interaction with dynamic environments. Unlike supervised learning, which relies on static labelled datasets, RL involves an agent learning to select actions that maximise cumulative rewards, making it particularly well-suited for contexts in which financial decisions (e.g., credit limit management, portfolio allocation, order execution) unfold over time with feedback loops [40]. Early applications demonstrated RL's potential in adapting strategies to shifting market conditions, such as dynamic portfolio rebalancing by learning from direct interaction with simulated or historical price data [41] and optimising trade execution to minimise market impact [41, 42]. More recently, RL has been used for credit risk assessment [43], dynamic pricing of customer credit [44], and intelligent decision-making [45]. These approaches leverage RL's capacity to handle partially observable, non-stationary markets, where feedback loops and delayed rewards are inherent. More recent work has extended RL beyond single-agent settings, exploring how autonomous trading bots or robo-advisors might cooperate or compete within complex market environments [46].

Multi-Agent Systems: MAS address scenarios where multiple autonomous agents interact whether in coordination or competition. From an algorithmic standpoint, MAS approaches often draw on game-theoretic principles or distributed computing to facilitate real-time negotiation, bidding, or consensus. Furthermore, MAS primarily focus on designing algorithmic solutions for autonomous decision-making entities that can operate in real-world environments, with emphasis on agent architecture, communication protocols, and coordination mechanisms. As an example, Tesauro and Bredin [47] employed MAS frameworks to model auction markets in which agents strategically bid under incomplete information, highlighting the adaptive complexity of competitive trading environments. Recent explorations of decentralised finance (DeFi) protocols further exemplify MAS setups, where incentive-aligned agents manage liquidity or execute trades without central intermediaries [48]. Although these systems can dynamically balance competing objectives such as profit maximisation, liquidity maintenance, or risk mitigation, they also require rigorous equilibrium analyses to avoid destabilising feedback loops [49]; (see Axtell and Farmer [50] for a detailed review).

Agent-Based Modelling: In contrast to MAS, which emphasise algorithmic solutions for coordinating or competing agents, ABM focuses on simulation to study emergent phenomena arising from interactions among agents [51]. While both ABM and MAS could involve multiple interacting agents, ABM is primarily a simulation methodology used to understand complex systems through computational experiments, rather than developing deployable autonomous systems to optimise for complex challenges. ABM represents actors in a system as individual agents with simplified actions, with the objective of understanding emergent behaviour from the aggregate of these simplified agents, in contrast to the more complex architectures used by agents in MAS to solve real-world optimisation issues, like auction markets [47]. ABM typically uses simplified agent representations of actors in system to model emergent behaviors, whereas MAS often implement more sophisticated agent architectures intended for real-world deployment.

ABM can be used to construct synthetic environments where varied agent archetypes such as customers, technical traders, and institutional players interact under user-defined rules [52]. Through these interactions, ABM sheds light on emergent effects like systemic risk, response to scenarios, or flash crashes [53]. In the banking sector, studies have famously used ABM to get insights on customer response to credit card promotions [8], for understanding the potential effects of bank behaviour on financial stability [54], to investigate diffusion of mobile-based branchless banking services [55], and to assess the effects of regulatory caps in the loan-to-value (LTV) ratio for housing mortgages [56]. Central banks and regulatory institutions have also employed ABM to stress-test monetary policies, examining how contagion or herding behaviours ripple across interconnected markets (see Aymanns et al. [57] for details). Historically, the reliance of ABM on stylised assumptions limited its predictive precision; however, in recent times, the fidelity of agent-based models to real-life counterparts has received greater attention, leading to the development of a range of approaches for model calibration [58] and validation [59].

As financial systems become increasingly adaptive and interconnected, the complementary use of RL, MAS, and ABM holds promise for capturing the complex, emergent dynamics that simpler modelling paradigms often overlook. Agents are core elements of these paradigms, which are reviewed next.

2.2 Overview of Agents

We align with the definition of an *agent* as a computational entity situated within an environment, capable of perceiving (and analysing) that environment and taking actions to achieve designated objectives independently or in interplay with other agents [60, 61]. In essence, an agent is able to operate without constant human oversight, making context-aware decisions in real time based on incoming signals or observations. Within the financial domain, this can manifest as a trading algorithm that monitors market data and executes buy-sell orders, or a robo-advisor that interprets client requirements and offers personalized budgeting advice. Crucially, agents may exhibit varying degrees of sophistication from simple programs reacting to predefined triggers, to adaptive, learning-based systems that refine their strategies in response to changing market or regulatory conditions.

To systematically characterise these computational entities, research often employs a taxonomy of agentic capabilities, covering dimensions such as:

- **Autonomous vs. Assistive**: An autonomous agent independently pursues its goals (e.g., a high-frequency trading bot adjusting to intraday volatility), whereas an assistive agent works alongside human operators (humans-in-the-loop), providing suggestions or automating discrete tasks (e.g., an email writing agent providing email content suggestions) [62].

- **Single vs. Multi-Agent**: Single-agent systems address standalone tasks, like portfolio rebalancing, while multi-agent environments involve multiple agents that may coordinate or compete, exemplified by multi-party auctions [63].
- **Reactive vs. Proactive**: Reactive agents respond primarily to immediate stimuli (e.g., event-driven alerts), whereas proactive agents anticipate future conditions, scheduling tasks and formulating long-term strategies [64].
- **Stateless vs. Stateful**: Agents may be stateless reacting only to current inputs or stateful, retaining historical information or learned parameters to guide decision-making over time [65].

Each of these dimensions plays a pivotal role in determining an agent's role, complexity, and operational requirements in financial systems, shaping not only the agent's decision-making processes but also the ethical, regulatory, and risk considerations associated with its deployment. Next, we provide an overview of LLMs, which are vital for significantly enhancing the capabilities and applications of such agents, particularly within the financial domain.. LLMs offer unprecedented natural language understanding and generation abilities that can enhance how agents perceive, reason about, and interact with complex financial environments.

2.3 Overview of Large Language Models

Large language models are a class of deep learning architectures designed to understand, generate, and manipulate human language at scale. Built on the transformer architecture introduced by Vaswani [66], LLMs leverage self-attention mechanisms to process sequential data in parallel, enabling efficient training on vast corpora of text. Models like OpenAI's Generative Pre-trained Transformer (GPT) series [67, 68], Meta's LLaMA [69], Anthropic's Claude [70], more recently DeepSeek [71], and domain-specific variants such as BloombergGPT [72] utilise layered transformer blocks to capture syntactic, semantic, and contextual relationships in text. In finance, encoder-decoder architectures (e.g., T5; Raffel et al. [73]) and retrieval-augmented generation (RAG) techniques are particularly relevant for tasks requiring both comprehension and generation, such as synthesising regulatory documents or generating earnings summaries. These models pre-train on diverse datasets such as financial filings, news articles, and earnings call transcripts to learn domain-specific linguistic patterns, then fine-tune on narrower tasks like sentiment analysis or anomaly detection [74].

2.3.1 The Importance of LLM-powered Agents for Banking and Finance

As mentioned in the earlier Section 1, LLM-powered agents represent a paradigm shift in financial AI by integrating linguistic reasoning with decision-making autonomy. These agents are uniquely critical to banking and finance due to the sector's growing reliance on unstructured data, demand for hyper-personalization, and need for scalable compliance amid tightening regulations. The key challenges that they can address in banking and finance are as below:

- **Unstructured data challenges**: Financial services increasingly hinge on interpreting unstructured text such as earnings calls, regulatory filings, and social media sentiment, where traditional models falter due to their reliance on structured inputs. LLM agents can bridge this gap by extracting actionable insights from linguistic data, enabling institutions to automate tasks like real-time fraud detection in transaction narratives [75] or anomaly detection in stock values [76].

- **Hyper-personalisation**: As customer expectations shift toward personalized services (e.g., tailored wealth and budget management or dynamic credit offers), LLM agents can empower hyper-personalisation by analysing individual financial histories, risk profiles, and behavioural patterns [77].
- **Regulatory scalability**: The sector's regulatory complexity, exemplified by evolving frameworks like the EU AI Act and Basel III, demands adaptive systems that can parse legal texts, flag non-compliant activities, and generate audit trails, tasks where LLMs can excel [78].
- **Real-time decision making**: LLM agents can address the need for real-time scalability in high-frequency trading and liquidity management, where their ability to process multilingual news feeds and macroeconomic signals can reduce latency in decision-making [48].
- **Contextual awareness and memory**: While rule-based systems and shallow NLP tools lacked contextual awareness and memory, LLM agents can infer intent from ambiguous queries and can retain memory from ongoing conversation (e.g., contextualising for the original customer query despite being furnished new details).
- **Systemic risk modelling**: LLM agents can also enhance systemic resilience by simulating emergent risks, such as bank runs or cryptocurrency crashes, in agent-based models, outperforming traditional equilibrium-based approaches [79].
- **Generative capabilities**: LLM agents possess generative capabilities, such as drafting responses to customer emails and drafting investment memos, which traditional models cannot replicate with as much finesse and consistency [80].
- **Multilingual and multimodal capabilities**: The ability of LLMs (and as a result of LLM agents) to process multilingual and multimodal data (e.g., text paired with tables) positions them as versatile tools for globalised financial operations.

2.3.2 Limitations of LLMs

Despite their promise, LLMs face significant limitations that have downstream effects on LLM agents, especially in the financial context. Although Section 4 details the challenges and limitations of LLM agents, it is crucial to note that LLMs struggle with precise numerical reasoning – often producing approximations or arithmetic errors that can jeopardise applications like portfolio optimization, derivative pricing, and risk modelling, where even minor miscalculations may lead to significant financial losses [81]. Additionally, LLMs can generate hallucinations by producing plausible yet factually incorrect outputs when extrapolating beyond their training data, a risk that could be catastrophic if errors appear in regulatory filings, audit reports, or trading signals [82].

Ethical concerns also arise from biases in training data that may result in unfair loan approvals or discriminatory credit scoring, while the opaque, "black-box" nature of LLMs conflicts with regulatory demands for explainability [83]. In fact, recent investigation into LLM biases towards certain nationalities and culturally-indicative names highlights that LLMs frequently default to stereotypical associations, thereby raising concerns about real-world applications [84]. Moreover, fixed token limits may truncate vital context in lengthy financial documents, and security vulnerabilities such as susceptibility to prompt injection attacks pose systemic risks by potentially exposing sensitive customer information [85, 86]. Last but not least, studies have found that LLMs are too rational and lack fidelity to human behaviour [87–89]. Mimicking human biases and accounting for human irrationality in financial decisioning is necessary for LLM applications in finance and banking, since financial decision-making is inherently behavioural and LLMs need to align with, and not override, human psychology to avoid systemic, ethical, and operational risks.

2.3.3 Regulatory, Compliance, and Legal Constraints

Even apart from the operational limitations concerning the use of LLMs, the ever-changing dynamics in the field, as well as regulatory, compliance, and legal constraints make it challenging for mission-critical sectors to deploy these models. These constraints are listed as follows:

- **Rapid evolution of LLMs**: The rapid evolution of LLMs, marked by frequent releases of open-source frameworks (e.g., LLaMA-3, DeepSeek-R1), proprietary advancements (e.g., ChatGPT o1-pro and o3 models, Qwen, Claude), and specialized financial tools (e.g., BloombergGPT), has outpaced the development of robust regulatory frameworks.

- **Shifting partnerships among LLM suppliers**: The transient nature of LLM suppliers, evidenced by shifting partnerships (e.g., Microsoft-OpenAI, Anthropic-Google) and volatile open-source ecosystems, further complicates long-term compliance.

- **Evolving monitoring needs and lack of transparency**: The open-sourcing of models such as Meta's LLaMA series has raised concerns about unmonitored deployment in sensitive workflows [80], while the lack of transparency in third-party LLM providers complicates audits for bias, security, and alignment with financial regulations [90]. Moreover, LLMs' "black-box" nature conflicts with regulations like GDPR's "right to explanation" and Basel III's model risk management requirements [91].

- **Regular feature releases**: Vendors are regularly releasing new functionalities such as robotic process automation (RPA) with LLMs [92], like enabling LLMs to control the user's computer screen and web-browser (e.g., Anthropic's Claude 3.5 Sonnet and OpenAI's Operator Assistant). Such features are marred with vulnerabilities like prompt-injection attacks, wherein malicious text presented to the LLM on screen can override user instructions, presenting a significant security vulnerability in these systems.

- **Data provenance**: Training on unvetted internet data risks embedding biases or copyrighted material, violating anti-discrimination laws (e.g., U.S. ECOA) and intellectual property norms [83, 93].

Compliance strategies: As mentioned, the growth in the field of LLMs has outpaced the development of robust regulatory frameworks. This creates a dynamic tension between innovation and compliance, particularly in highly regulated sectors like finance and banking. Regulatory bodies, such as the European Union under its AI Act, classify high-risk AI systems (e.g., credit scoring, fraud detection) as requiring stringent oversight, including transparency, human oversight, and robustness assessments [94]. LLM applications in finance, particularly those influencing customer outcomes (e.g., robo-advisory, loan underwriting), are likely to fall under these high-risk categories, requiring rigorous documentation of training data, model behaviour, and decision logic [78]. Financial institutions must navigate through fragmented standards, such as the U.S. SEC's focus on AI-driven conflicts of interest [95] and the Bank of England's warnings about systemic risks from generative AI [96]. To mitigate these risks, institutions are adopting hybrid governance models, such as human-in-the-loop validation for high-stakes decisions [97] and synthetic data pipelines to audit model behaviour [98]. However, as the Financial Stability Board (FSB) notes, global coordination remains critical to harmonise standards and prevent regulatory arbitrage [99].

3 Landscape of LLM Agents in Banking and Finance

Having contextualised the field, we now provide a survey of the different works that have explored the use of LLM agents across a range of applications in banking and finance. We have classified these applications by different agent functions, namely *simulation, analysis, advising*, and *acting*. While anchored in the functional taxonomy, the review covers agent applications across subject domains (retail banking, investment banking, macroeconomic forecasting, etc.) and architectural frameworks (single- vs. multi-agent), providing a holistic view of the LLM-agent landscape in the sector.

3.1 LLM Agents in Simulation

Simulation is a popular tool for modelling uncertainties, understanding complex system dynamics, optimising strategies, and enhancing practical understanding of market behaviours in a dynamic, controlled environment (or 'sandbox') that traditional analytical methods cannot easily capture [51–53]. By allowing experiments with multiple "what-if" scenarios, simulations help identify critical risk factors and key leverage points, thereby guiding better strategic decisions.

In the previous section, we summarised how simulation through traditional ABM in various aspects of finance and banking – from pricing to operational planning and risk governance – is helping institutions make more informed decisions, enhancing both their resilience and competitive edge. However, traditional ABM comes with challenges such as reliance on hand-crafted rules that can be brittle in dynamic environments [100] and limited contextual understanding in complex environments [101] that LLM agents can help resolve. In this section, we review the studies that have used LLM-powered agents in financial simulations, replacing or complementing traditional (e.g., rule-based) computational agents. Leveraging their linguistic and advanced reasoning capabilities, LLM-powered agents have the potential to address key design and computational challenges of traditional simulations. Their ability to generate contextually appropriate and diverse responses as well as to assume specific roles or personas enables more nuanced and realistic agent interactions in simulations. These richer simulations have a great potential in a variety of financial settings, including macroeconomics, trading, and retail banking and have been adopted in other fields, including the social sciences [102], medicine [103], and life sciences [104, 105].

3.1.1 Market Simulation

Several works have explored the use of LLM agents in simulating markets, finding that LLMs can effectively act as surrogates for humans, albeit with an unclear outlook on when they outperform traditional trading strategies.

Auction simulation. An environment for evaluating LLMs' strategic abilities in competitive auction settings, named AUCARENA, is introduced by Chen et al. [106]. The authors use the environment to evaluate the performance of agents powered by LLMs (including GPT-3.5, GPT-4, Mistral, Mixtral, and Gemini) in resource management, risk evaluation, and adaptability in a series of auction-based experiments. Their analysis reveals that while LLMs like GPT-4 demonstrate superior performance in most auction scenarios due to their adaptability and strategic foresight, they are occasionally outperformed by simpler, rule-based agents.

Similarly, Guo et al. [107] introduce the EconArena framework to run behavioural economic games with LLM agents to investigate their ability to adapt and learn from past experiences in partially observable auction game simulations. LLM agent behaviour is examined under different game configurations, player setups, and amounts of historical game knowledge provided to the agents. They find that LLMs perform best when informed of their opponents' strategies and given access to game history, but behave too rationally without game history. However, the study does not make the LLMs compete directly against each other, which would be an interesting evaluation for future studies.

Trading simulation. Several works have also created LLM agent systems for traditional financial trading. Among these, Li et al. [108] developed a multi-agent system called TradingGPT, which incorporates a layered memory structure and distinct trading characteristic profiles to enhance financial trading performance. The framework aims to improve decision-making in financial trading by managing and prioritising information efficiently across its layered short, medium, and long-term memory system, where macro-level indicators are stored in long-term memory and daily investment messages in short-term. Additionally, by allowing for multiple agents with varying risk preferences, the framework also enables increased decision-making diversity, delineating a roadmap for future trading simulation frameworks to address the widely popular shortcomings of underlying LLMs such as catastrophic forgetting and homogenous decision-making [109].

In a similar vein, the work of Gao et al. [110] introduces an Agent-based Simulated Financial Market (ASFM) framework in a simulated stock market with an order-matching system mirroring real-world financial markets. The work uses LLM-based agents to allow for the incorporation of complex human behaviours, such as behavioural biases and differing investment strategies to simulate the real stock market's response to economic regulatory policies. The study tests LLM agent response to economic shocks like interest rate changes and inflation, thereby providing an approach for policymakers to simulate and analyse the potential impacts of their decisions (such as rate cuts) on financial markets. Overall, the study finds that the agents display a sophisticated understanding of market dynamics, producing results consistent with traditional economic theories.

The work of Zhang et al. [111] introduces StockAgent, a multi-agent LLM framework for trading stocks. The work explored the difference in behaviours of Gemini-driven and GPT-driven agents as investors while accounting for the impact of various external factors on stock trading activities. The paper also claims to address the test set leakage issues present in existing AI-based trading simulation systems, where the leakage due to underlying training data used for training LLMs can bias LLM agent response to stimuli. This work noted that GPT-driven LLM agents trade less frequently but with higher volume than Gemini-driven agents when presented with the same decisions, guiding the design of more robust and reliable agentic trading systems by incorporating comprehensive external factors and addressing the biases of different LLMs.

3.1.2 Macro and Micro-Economic Simulation

Macro-economic simulation. Macro-economic systems have been studied extensively through agent-based modelling [112–114], making the introduction of LLM agents an exciting approach to address some of the challenges that are present in agent-based modelling. This potential is shown in the work of Li et al. [115], who present the simulation of an economy with several market dynamics and an LLM-powered agent representing individual people, named EconAgent. The agents are allocated certain age and income from historical distributions and are equipped with perception and memory modules to account for previous months' calculations (such as government taxation, price of goods, and an interest rate from a central bank). Evaluating the fidelity of their simulation, the authors find that the EconAgent produces more stable and plausible macroeconomic indicators compared to the traditional ABM approach, correctly replicates established economic relationships, and shows interpretable decision-making aligned with well-known economic theories like Phillips Curve and Okun's Law, providing a solid argument in favour of using LLM agents for macroeconomic simulations.

Douglas and Verstyuk [116] integrate LLM agents into a canonical consumption-savings framework [117] to simulate more realistic human decision-making under economic uncertainty. The study uses detailed prompts providing economic context and specific decision-making questions, and includes a version where agents can communicate based on U.S. input-output network data. Compared to the original model and a constrained-information baseline, economies with LLM agents show higher wealth inequality, more pronounced economic fluctuations, and greater volatility in key economic indicators. LLM agents tend to be more frugal, especially when unemployed, leading to higher overall wealth accumulation. When allowed to communicate, these agents contribute to even higher aggregate wealth and output, highlighting the impact of information sharing on economic decisions.

In another example of macroeconomic simulation, while not directly using LLM agents, Dwarakanath et al. [118] have built ABIDES-Economist, a multi-agent simulator for economic systems incorporating heterogeneous agents such as households, firms, a central bank, and a government. The simulator builds off of ABIDES [119], a widely used simulator of financial markets, which allows these agents to interact in a simulated economy, with the ability to implement both rule-based strategies and reinforcement learning (RL) techniques. The ABIDES-Economist platform demonstrates that RL can effectively be used to model and simulate complex economic behaviours, allowing for the exploration of dynamic scenarios that traditional economic models may struggle to capture to better answer policy questions. Potential future work can include substitution of RL-agents in the framework with LLM agents.

Micro-economic simulation (retail and customer). A number of works have found that LLM agents are able to take on clear roles as buyers and sellers. For example, one of the earlier works using LLM agents for simulation is by Nascimento et al. [120] that simulates an online book marketplace environment. The buyers and sellers, who are LLM agents, interact within this virtual marketplace. Buyer agents aim to purchase a single book at the lowest possible price, while seller agents, possessing identical books, have the freedom to set their own prices but lose the deal by the third negotiation. This competitive setup creates a free-market scenario where successful seller agents maximise profits and successful buyer agents minimise costs. During their simulation, the authors find that the LLM agents are sophisticated in their communication, possess adaptability in dynamic environments, and have robust problem-solving capabilities.

Subsequent studies used these micro-economic simulations to explicitly see if LLMs can replicate known economic phenomena. For example, a virtual town with restaurant and customer agents powered by LLMs is developed by Zhao et al. [121], where the restaurant agents compete among each other to attact customers and this competition encourages them to transform, such as cultivating better operating strategies. Several experiments reveal that well-known market and sociological theories are replicated in this simulation, including the imitation of product promotions, improvements to product quality under competition, and the Matthews' effect.

Among other simulations testing the adaptive pricing capabilities of LLM agents, the work of Han et al. [122], who develop a simulation of "smart agents" powered by GPT-4 that represent firms, and interact with one another in a canonical Bertrand duopoly setting to study firm price competition and collusion behaviours. The study's findings reveal that smart agents consistently achieve tacit collusion without communication, resulting in prices above the Bertrand equilibrium but below monopoly or cartel levels. With communication allowed, agents reach higher-level collusion near cartel prices. In sum, communication among LLM agents facilitates faster collusion formation and encourages small price deviations to explore win-win scenarios, reducing price war risks.

The work of Wu et al. [101] very similarly explores how LLMs can be utilised to model firms in a simulation. The work also finds that LLMs are able to exhibit adaptive pricing strategies, responding to market conditions and competitor actions, and are also able to engage in more complex behaviour that requires coordination and communication, including collusion to adjust prices in a duopoly. This work supplements the findings of Han et al. [122] that LLM agents are able to take on different roles, adjust pricing strategies effectively, and incorporate conversations and debates into their decision-making.

Finally, the effect of information economics is explored by Weiss et al. [123], who explored the behaviour of LLM-powered agents in a simulated digital marketplace. They find that these agents can effectively inspect information and make purchasing decisions and find that higher-quality outcomes result whenever inspector of goods occurs.

3.1.3 Synthetic Data Generation

Since financial data is often sensitive, proprietary, or limited in availability, synthetic data offers a valuable alternative for training and testing various analytical models. LLM-powered agents have the potential to generate synthetic data that is realistic with human-like decision-making patterns in a controlled environment. Researchers can then use this data to explore various policy scenarios, test theoretical models, and conduct robust "what-if" analyses without the limitations imposed by scarce or sensitive real-world data. While synthetic data generation using LLMs is common in non-finance related tasks (see Abdullin et al. [124], Sengupta et al. [125], Long et al. [126]), the work of Zuo et al. [127] introduces an innovative paradigm to generate synthetic financial data. In this paper, the authors use an RL-based *"Selector"* agent to generate a prompt upon selecting keywords for the data to be generated, which is then passed to an LLM-powered *"Executor"* agent that produces this synthetic financial data, while maintaining data privacy. Their experimental results show that models trained on their synthetically generated data perform competitively compared to those trained on actual financial data. Cai [128] also make use of FinGPT to generate synthetic datasets for credit risk assessment. They then compare the performance of original and FinGPT-generated datasets on traditional machine learning algorithms, finding that FinGPT-generated datasets can enhance the predictive accuracy of these models, particularly in scenarios where data is scarce. While the research on synthetic financial data generation is limited, it is identified as an application of LLM agents with a huge potential [126].

3.2 LLM Agents as Analysts

LLM-powered agents possess the capability to rapidly process and distill vast amounts of complex data by acting as digital assistants, thereby augmenting human decision-making. In banking, these agents can generate detailed reports, perform real-time risk assessments, and support regulatory compliance by parsing unstructured data from market news, customer feedback, and financial statements. Similarly, in finance and economics, these models can integrate diverse data sources to identify trends and forecast market shifts, offering insights that help shape monetary policy and drive investment strategies. This assistive role would not only streamline routine tasks but also enable financial experts to focus on higher-level strategic decision-making and innovative problem-solving. In this section, we explore the diverse range of analytical applications where LLM agents have demonstrated potential.

3.2.1 Market and Sentiment Analysis

The surge in textual data - from market-moving social media posts[1] [2] to real-time news on geopolitical conflicts - has made LLM agents indispensable in market and sentiment analysis. Through these agents, we can now parse vast, unstructured datasets at scale, detecting nuanced sentiment shifts or geopolitical risks that drive market volatility. By analyzing linguistic subtleties and contextual cues, LLM agents can provide real-time insights, enabling traders and institutions to anticipate trends, mitigate risks, and capitalise on opportunities faster than traditional methods.

Market analysis. Among the first works in this area was that of Bybee [129], who used *"representative agents"* powered by GPT to find correlations between expected returns estimated by investors (through the Survey of Professional Forecasters, the American Association of Individual Investors, and the Duke CFO Survey) and those predicted by LLMs prompted with relevant headlines from Wall Street Journal. The author found that the expected returns calculated by the LLM agent and traditional methods (the survey counterparts) correlated, in addition to a significant correlation between the macroeconomic expectations of the LLM and the Survey of Professional Forecasters (SPF)[3], highlighting the ability of LLM agents in performing market analysis.

Similarly, the work of Wu [130] scored news headlines using three LLMs (GPT-3.5, Qianwen, and Baichuan) and observed correlation between these sentiment factors and market performance, suggesting that this was an effective strategy for forecasting, that market news is more influential than general news, and the market was more sensitive to negative news than positive news. In a similar work, Fatouros et al. [131] introduce MarketSenseAI, which integrates Chain of Thought (CoT) prompting and in-context learning into the stock selection of GPT-4 powered AI agents. They find that the stock selection of the S&P 100 generates strong returns across a range of trading metrics.

Sentiment analysis. In addition to other factors, market sentiment plays a key role in finance and trading. In banking, customer complaint sentiment analysis has helped banks improve their operations [132]. The capacity of LLMs to process and analyse extensive textual datasets has rendered sentiment analysis a prominent application of LLM agents within the financial sector. This synergy between LLMs' capabilities and the demands of sentiment analysis in finance and banking has led to widespread adoption and research interest in this domain. As an example, Zhang et al. [133] used LLMs to extract sentiment factors from news summaries, while the LLMFactor framework Wang et al. [134] employs Sequential Knowledge-Guided Prompting uses LLMs to extract features for time series analysis from news articles, with a focus on explainability. The work of Pfeifer and Marohl [135] introduces CentralBankRoBERTa, which is fine-tuned on central bank communications with the objective of identifying entities and sentiments in these texts. Similarly, Wu [130] uses LLMs to analyse news stories and highlights the sensitivity of market performance to the sentiment of news headlines, while Kirtac and Germano [11] explores the capabilities of LLMs like GPT and BERT for financial news sentiment, finding they outperform traditional dictionary models.

LLM frameworks that are more agentic have been used to enhance sentiment analysis. This includes the work of Ding et al. [136], who extract text-based features from news, policies, and trends to be used as a component in their framework for stock prediction. The work of Xing [137] introduces a multi-agent LLM

[1] https://www.theguardian.com/business/2021/feb/04/price-of-dogecoin-rises-by-50-following-elon-musk-tweet

[2] https://www.coindesk.com/markets/2024/10/16/dogecoin-price-surges-10-as-elon-musks-department-of-government-efficiency-gains-traction

[3] "The SPF is a quarterly survey of macroeconomic forecasts conducted by the Philadelphia Federal Reserve and remains the gold standard for work studying macroeconomic expectations"

framework for predicting the sentiment of financial news, motivated by the "Society of Mind" theory. The LLM agent frameworks of FinMem [12] and FinRobot [138] include document analysis as core components of their frameworks. The work of Phan [139] has used LLM agents to extract insights from news articles to power macro-economic analysis, improving the predictive power of supervised learning techniques like gradient-boosted trees (GBTs) for predicting macroeconomic indicators.

3.2.2 Stock Price Prediction

Effective stock price prediction necessitates leveraging both extensive historical data and recent market trends; historical data provides valuable context regarding long-term patterns and economic cycles, while recent data captures current market dynamics and investor sentiment, enabling more accurate forecasts and informed decision-making. Moreover, mimicking traditional investment strategy formulation, i.e., discussions between agents with different perspectives on a stock's future is also key. In light of this, Li et al. [108] developed a multi-agent system called TradingGPT which incorporates a layered memory structure and distinct trading characteristic agent profiles to enhance financial trading performance. By showing how multi-agent architectures can be built for the simulation of LLM agents for stock-trading scenarios, this method paper promises superior trading performance compared to other leading trading agent systems by simulating the cognitive behaviours of human traders and ensuring adaptability in a constantly evolving market environment.

Zhang et al. [111] designed StockAgent, a multi-agent AI system powered by LLMs to simulate investor trading behaviours in response to real stock market conditions such as policy changes, company fundmentals, global events, and other macroeconomic conditions. With StockAgent, the authors evaluated GPT- and Gemini-based agents, while also addressing the issue of "test data leakage", whereby previous stock prediction frameworks (inadvertently) leveraged prior knowledge of the LLMs to test the LLMs. On the question of trading behaviours in response to real stock market conditions, the authors found that higher reliability in the StockAgent simulation is achieved when comprehensive data corresponding to the real stock market conditions is provided. Additionally, they found that agents driven by different LLMs exhibit significant differences in trading behaviour, i.e., GPT-driven AI agents trade less frequently (fewer trades per trading round) but with higher volume per trading round compared to Gemini-driven agents, which show higher frequency of trading activity across different stocks.

The previously mentioned work of Gao et al. [110] introduce an Agent-based Simulated Financial Market (ASFM) framework that includes a simulated stock market with an order-matching system mirroring real-world financial markets. The work of Wang et al. [134] integrated background knowledge, stock-related factors, and temporal data to forecast Chinese and U.S. stock movements with LLMs, outperforming traditional forecasting techniques. The work of Cheng and Chin [140] introduces a hybrid machine learning approach for stock price prediction, utilising both quantitative factors from asset pricing with qualitative analysis of financial information from an LLM agent to predict future stock prices, finding similar or superior performance to other machine learning techniques.

In the work of Koa et al. [141], the authors introduce a Summarize-Explain-Predict (SEP) framework, which utilizes a verbal self-reflective agent and Proximal Policy Optimization (PPO), finding that it outperforms other deep learning and LLM-based techniques. Yang et al. [138] introduces the open-source AI agentic platform FinRobot. This platform consists of four major layers (or components): Financial AI Agents, Financial LLM Algorithms, LLMOps and DataOps, and Multi-source LLM Foundation Models, providing a holistic framework for developing financial AI agents.

The inclusion of different modes of data has also been explored, including in the work of Zhang et al. [142], who introduce FinAgent, a multimodal financial trading agent that integrates both textual and visual data. The authors compare their framework against a range of techniques, including the LLM-based techniques of FinGPT [143] and FinMem [144], finding that FinAgent broadly outperforms other techniques.

Multi-agent strategies have also been explored, with the work of Yu et al. [12] developing FinCon, a multi-agent multi-modal technique that can use a wide range of data sources, including financial documents, reports, and audio recordings. Comparison indicates that FinCon outperforms other techniques, include FinGPT [143], FinMem [144], and FinAgent [142]. These works exhibit a clear trend of improvements in performance as LLM-based agent frameworks are able to leverage an increasing variety of modes of data, with increasingly sophisticated agent architectures.

3.2.3 Document and Compliance Analysis

Document analysis. The advanced NLP capabilities of LLMs make LLM-powered agents an effective tool for extracting key insights from extensive reports and documents. An example of this is the open source financial AI agent platform, FinRobot, where Yang et al. [138] design an open-source AI agent platform to support multiple financially specialized AI agents powered by LLMs. The platform consists of four major layers: Financial AI Agents, Financial LLM Algorithms, LLMOps and DataOps, and multi-source LLM Foundation Models. FinRobot provides a holistic framework for developing financial AI agents capable of performing a wide range of tasks, from market forecasting to financial document analysis. The platform also introduces a Smart Scheduler mechanism that allows seamless integration of multi-source LLMs, enhancing performance by selecting the most suitable models for specific financial tasks. In other works, Wan et al. [145] examined the application of multi-agent AI frameworks to improve the review processes for underlying asset documents in structured finance. They evaluated the performance of their framework by cross-verifying information between a user's loan application and their bank statement finding that the use of dual-agent systems enhanced accuracy compared to a single agent, although at the expense of higher operational costs.

Legal, compliance, and customer due diligence. Maintaining compliance with regulation and legislation is of great importance for financial institutions. Works have shown how LLMs can enhance this in several ways, including applying regulations, verifying regulatory compliance, researching individuals and institutions, and assisting in compliance tasks. Berger et al. [146] explore the abilities of LLM agents in regulatory compliance verification, finding that LLM agents have the potential to accelerate regulatory compliance verification. Regarding the interpretation of financial regulation, Cao and Feinstein [147] investigate the use of LLMs to translate complex regulatory texts into actionable code. They find that models like GPT-4 are particularly effective at extracting relevant information and performing related calculations. Understanding the roles of customers, both individuals and institutions, is a key part of compliance and due diligence for banks, including identifying politically exposed persons. Works like Loffredo and Yun [148], which uses a RAG-powered LLM agent to support scholars understanding congress. This approach could potentially be applied to due diligence processes in banking and finance. The work of Licari et al. [149] investigates the abilities of LLM-powered agents labelled *"LLM Evaluators"* to improve the understanding of banking supervisory regulation, introducing a multi-step prompt construction method to improve the performance in question-answering tasks.

Blockchain and decentralised finance. Research in blockchain and decentralised finance with LLM agents as analysts has focussed on automating the analysis of smart contracts (decentralised applications built atop blockchains). Ma et al. [150], Wei et al. [151], and Xiao et al. [152] have demonstrated the efficacy of fine-tuned LLM agents in this domain. Ma et al. [150] introduced iAudit, an LLM agent framework inspired by human auditors that combines fine-tuning and LLM-based agents for smart contract auditing with justifications. Mimicking human auditors, their framework consists of fine-tuned "Detector" and "Reasoner" models that make decisions and generate causes of vulnerabilities in the contracts, followed by "Ranker" and "Critic" agents that iteratively select and debate the most suitable cause of vulnerability based on the output of the fine-tuned Reasoner model. The authors find their hybrid approach outperforms zero-shot LLM prompting and fine-tuned models. In a similar vein, Wei et al. [151] introduce FTSmartAudit, a framework with the same application as iAudit, with the key difference in the use of smaller, fine-tuned models to perform smart contract auditing. The authors find that even fine-tuning smaller LLMs can help detect vulnerabilites in smart contracts. Finally, in another closely related work, Xiao et al. [152] extended the application of LLM agents to the above application by extending the coverage to five LLMs and using advanced prompting. These studies collectively highlight the potential of LLM agents in enhancing the security and reliability of blockchain-based financial systems through automated contract analysis and vulnerability detection.

3.3 LLM Agents as Advisors

LLMs possess several attributes that render them particularly suitable for serving as advisors to both financial professionals and customers. Their capacity to assimilate vast amounts of financial information, coupled with their ability to assume specific roles through tailored prompts, enables them to address individual contexts and problem-solving scenarios effectively. Additionally, recently developed LLMs have exhibited strong abilities to make effective and informed decisions based on their world knowledge, the context provider, and the parameters for a problem provided to them through the prompts they receive [153, 154]. The ability to tune the behaviour of an LLM through prompting in natural language means LLMs can be calibrated to match the user's knowledge, and their conversational capabilities facilitate in-depth exploration of topics, allowing users to seek clarification as needed [155]. These abilities make it appealing for LLM agents to act as advisory assistants, providing a paradigm shift over classical machine learning approaches, such as supervised learning, which lack the ability to provide context-specific recommendations. While research into LLM agents as advisors is still in its early stages, with a considerable portion likely occurring within corporate environments, this field is expected to undergo rapid development in the coming years.

3.3.1 Automated Customer Service

The potential of AI chatbots for automated customer service in banking has been highlighted by Birch and Rutter [156], who speculate that the rapidly growing ability and availability of LLMs could easily lead to many financial activities being taken over by intelligent agents. One of the earlier works exploring the evaluation of LLM-based chatbots in personal finance found that, compared to rule-based systems, LLMs could generate fluent responses, but often lacked the reliability and consistency necessary for sound financial advisement [157]. Advancing this field through more consistent evaluation, Yang et al. [158] introduces IDEA-FinBench, a benchmark for evaluating financial knowledge in LLMs and LLM agents. Alongside this benchmark, they presented several few-shot prompting and fine-tuning based frameworks for developing LLMs with enhanced financial knowledge. Furthermore, they proposed IDEA-FinQA, a multi-LLM agent framework designed for financial question answering, with its responses outperforming LLMs in terms of factuality, relevance, and informativeness.

3.3.2 Financial Planning and Budgeting

The potential of LLM agents for individual and household financial planning has been directly explored by de Zarzà et al. [159]. Their research proposes an optimisation framework capable of managing multiple incomes and expenditures, with an LLM agent providing tailored recommendations. Notably, the authors report that the generated financial plans are not only economically sound but also align with the financial goals and preferences of the target audience. While this study represents one of the few works addressing personal financial planning advice using LLM agents, it demonstrates the significant potential of this application.

3.4 LLM Agents as Actors

As mentioned in earlier sections, the banking and finance sector is under pressure to provide speedy, scalable, and precise solutions in an increasingly complex and data-driven landscape. Traditional systems are constrained by limitations of human capacity and manual workflows, falling short of delivering 24/7 availability, personalisation at scale, and high-frequency actions [160, 161]. Unlike traditional automation, which focuses on predefined action patterns, LLM agents exhibit autonomy that enables them to select appropriate actions based on their goals and environmental context, thus expanding the range of possibilities beyond conventional automation approaches [162]. As research progresses and implementation becomes more common, autonomous LLM agents are poised to significantly transform the landscape of financial services and banking operations. Recent developments, such as reasoning LLMs[4][5] and agents capable of direct computer interaction,[6][7] offer promising new opportunities in this field. While the critical nature of the sector raises skepticism over the autonomy of advanced agents in financial decision-making, there are areas in automated trading, money management, and customer interaction that have warranted exploration of the potential of these agents as actors, which we discuss in this section.

3.4.1 LLM-based Trading Strategies

In Section 3.2, we discussed the analytical capabilities of LLM agents in facilitating automated trading. Studies cited in this section further advance LLM agent capabilities, demonstrating how LLM agents leverage analytical findings to execute actions autonomously. The work of Wang et al. [163] introduces Alpha-GPT, an LLM agent-based system capable of autonomously analysing data, patterns, and market conditions to uncover strategies, assets, and trades that can outperform market on a risk-adjusted basis. This agent translates users' ideas into prompts for an LLM, which then generates valid alpha expressions along with explanations of the reasoning behind the choices.

Wang et al. [164] introduce QuantAgent, an LLM-based trading agent that utilises a domain-specific knowledge base to autonomously mine trading signals that is capable of self-improvement based on its performance. In a more complex approach to strategy selection in automated trading, Kou et al. [165] propose a novel framework for quantitative trading using a multi-agent system, where through sequential modules, the agents autonomously extract predictive signals by integrating numerical data, research papers, and visual charts, followed by constructing a diverse pool of trading agents with varying risk preferences, and finally employing a dynamic weight-gating mechanism to select and assign weights to the most relevant agents

[4]https://openai.com/index/openai-o3-mini/

[5]https://api-docs.deepseek.com/news/news250120

[6]https://docs.anthropic.com/en/docs/build-with-claude/computer-use

[7]https://openai.com/index/introducing-operator/

based on real-time market conditions for adaptive strategy selection. Li et al. [144] developed FinMem, an LLM-agent-based framework for financial trading, consisting of an agent with defined characteristics in its profile module, alongside memory and decision-making modules. Notably, this framework outperformed several other strategies, including non-specialized generative agents [13], FinGPT [143], PPO, and deep Q-learning.

The work of Zhang et al. [111] introduces a simulation of investor trading, with individual agents powered by LLMs, illustrating the potential of LLM agents as actors in various roles within a stock trading environment. Advancements in explainable AI have also been made in the context of stock predictions with LLM agents. For example, Koa et al. [141] introduce a SEP framework, which employs a verbal self-reflective agent and PPO. This approach outperformed other deep learning and LLM-based techniques, highlighting the ability for explainability in financial decision-making with LLM agents.

The exploration of multi-agent strategies has further expanded the capabilities of LLM agents in automated trading. Yu et al. [12] developed FinCon, a multi-agent multi-modal technique capable of utilizing a wide range of data sources, including financial documents, reports, and audio recordings. Comparative studies across a range of bearish and bullish stocks demonstrates that FinCon outperforms other techniques, including FinGPT [143], FinMem [142], and FinAgent [142].

3.4.2 Personal Financial Management

Several works have used LLM agents to enhance the ability of individuals' money to be managed, with indications that LLM agents are able to be effectively adapted to the needs of a wide range of financial customers.

The potential of LLMs in personal money management is demonstrated by de Zarzà et al. [159], who develop an LLM agent that is capable of providing financial recommendations "that are both economically sound... and aligned with the financial goals and preferences of the concerned parties". An early example of practical implementation in this domain is the personalised assistant for digital banking introduced by Easin et al. [166]. Their framework of multiple LLM agents demonstrates the capability to solve a range of financial tasks in an autonomous manner, showcasing the potential of these techniques in personal finance management. Simultaneously, the potential risks and regulatory challenges associated with such applications have garnered attention. Notably, Phillips [167] argues that regulatory bodies are currently ill-equipped to address the risks posed by LLM agents acting on behalf of individuals in financial matters.

3.4.3 Asset Pricing

Many researchers have explored the use of LLM agents for automated stock pricing. The work of Yang et al. [138] introduces an open-source platform for using LLM agents with a wide range of tasks, including stock price prediction. However, this work acknowledges a lack of integration with new incoming data. More recently, the work of Han et al. [168] introduces a heterogeneous multi-agent framework to assist in financial research. Experiments performed in this work indicate this multi-agent framework outperforms a single agent and highlights the potential for transformation in financial analysis. Broadly, the number of studies exploring the pricing and trading of options and derivatives are limited, although the potential is clear. For example, the work of Rouzel et al. [169] proposes an LLM-powered protocol for derivatives and options trading in the decentralised finance space.

3.4.4 Blockchain and Decentralised Finance

An LLM-based cryptocurrency trading agent, named CryptoTrade, is introduced by Li et al. [170], which uses both price features and cryptocurrency news to trade different cryptocurrencies. The authors find that this agent outperforms other trading strategies and is able to take a "buy the rumour, sell the story" approach to using text information in trading. Decentralised technologies have also been explored to facilitate the coordination of LLM agents.

The work of Kim [171] proposed a framework for providing an overview of global LLM agent activity, built on the blockchain, named Ethereum AI Agent Coordinator (EAAC). the applications of LLM agents for decentralised finance have mainly focussed on assessing the vulnerabilities of smart contracts and automated trading of cryptocurrency.

Another notable contribution to the field is the work of Li et al. [172], which introduces InvestorBench and evaluates the performance of a range of LLM and LLM agents [138, 142, 144] on cryptocurrency trading among other applications. The authors find wide variations in the abilities of LLM agents across stock, cryptocurrency, and exchange-traded fund (ETF) trading, and highlight the importance of choosing the most appropriate model for each task.

3.4.5 Personalised Customer Interaction

Ensuring current customers are able to effectively use financial products offered and being able to target potential new customers allows a financial institution to deliver positive outcomes for its customer base and maintain a competitive edge. The abilities of LLMs to tackle natural language problems and the ability for them to have their capabilities fine-tuned through natural language meant LLM agents are rapidly changing how organisations interact with current and potential customers.

Customer Service. The proliferation of AI chatbots in customer service predates the widespread adoption of LLMs for natural language processing tasks [173, 174]. With the advent of LLMs, their application in banking has expanded, particularly in the domain of question-answering systems. For instance, Kulkarni et al. [175] demonstrate the implementation of an LLM agent that uses Retrieval-Augmented Generation (RAG) for addressing customer inquiries. Similarly, Avagyan and Davtyan [176] present a RAG-based LLM agent leveraging Langchain to provide comprehensive assistance to bank customers.

Targeted marketing. The application of NLP in banking marketing has historically encompassed various aspects, including customer retention strategies, competitor product analysis, and rapid insight extraction from feedback [177]. LLMs have emerged as a potentially transformative tool in marketing [178], with widespread adoption across multiple sectors for diverse applications such as content creation, personalized customer engagement, and automated market trend research [179].

Several financial institutions have reported leveraging LLM agents to enhance their marketing efforts.[8][9] However, there appears to be a lack of published academic research specifically addressing targeted marketing with LLM agents in the banking and finance sectors.

[8] https://www.linkedin.com/pulse/15-ways-we-using-chatgpt-banking-chris-nichols/
[9] https://www.marketingdive.com/news/ally-bank-generative-AI-LLM-marketing-enterprise-workflows/700138/

3.5 Emerging Applications of LLM Agents

Several areas of finance and banking have not had many research works exploring the potential of LLM agents, despite the effectiveness they have shown in other domains. These areas have the potential to be transformed by investigation into the abilities LLM agents and are ripe opportunities for future investigation. We now highlight these areas, including current adjacent research in these domains.

3.5.1 Customer Wealth and Relationship Management

The potential of LLM-powered chatbots in customer relationship management (CRM) has garnered considerable attention in recent literature [180, 181]. Furthermore, the development of specific benchmarks for evaluating LLM agents in CRM applications, such as CRM-Arena [182] and the Salesforce AI Research CRM benchmark,[10], underscores the growing interest in this field. The integration of LLM agents in CRM represents a new frontier, as evidenced by Bank of America's Erica system, which has facilitated over 2 billion customer interactions,[11]. However, there appears to be a lack of published research on the application of LLM agents in CRM within the banking and financial sectors, possibly attributable to the proprietary nature of such developments within financial institutions, which may be reticent to disclose their research findings publicly.

Wealth advisory and management services [183], including estate planning, retirement strategies, and tax optimisation, align well with LLMs' ability to process complex data and solve intricate problems. Interest in using LLM agents for wealth management, with offerings from some large organisations[12][13] Additionally, Morgan Stanley has announced that LLM agents have been used to assist wealth managers.[14] However, there appears to be little academic research exploring the effectiveness of LLMs, or investigating the abilities of LLMs agents, in this field. This gap between industry adoption and academic study highlights the need for more research in this area. Future studies could focus on experimental evaluations of various LLM agent configurations, assessing their performance across different wealth management and advisory tasks.

3.5.2 Anomaly and Fraud Detection

While fraud detection has been a popular application for supervised machine learning for financial institutions [184, 185] and LLMs have been used for anomaly detection [186] in other fields, there has been limited application of LLM agents for these tasks in finance and banking. An LLM-based multi-agent framework for anomaly detection in the stock market is introduced in the work of Park [76], with each agent having a distinct role in the identification of anomalies in S&P 500 data. While this work does not directly evaluate the model performance, limiting the utility of this work, it shows the potential that multi-agent systems have for a range of financial tasks.

Chakraborty et al. [75] introduces a benchmarking suite for detecting and mitigating abuse and fraud, encompassing tasks such as toxic language identification and spam email detection. Their exploration in the abilities of LLMs indicate effectiveness in most tasks, with the exception of those requiring extensive rea-

[10] https://www.salesforceairesearch.com/crm-benchmark

[11] https://newsroom.bankofamerica.com/content/newsroom/press-releases/2024/04/bofa-s-erica-surpasses-2-billion-interactions–helping-42-millio.html

[12] https://www.alizila.com/ant-unveils-financial-large-language-model-llm-finance/

[13] https://www.benzinga.com/fintech/24/08/40294269/jpmorgan-rolls-out-ai-assistant-to-60-000-employees-for-enhanced-productivity

[14] https://www.morganstanley.com/press-releases/ai-at-morgan-stanley-debrief-launch

soning, but this appears to be the current limit of research for LLMs in fraudulent activity. Advancing the current literature, LLM agents could be employed not only to detect fraudulent activities but also to assist customers in navigating and mitigating fraudulent interactions, highlighting an opportunity for further investigation in this field.

3.5.3 Risk and Scenario Analysis

The application of LLMs in credit decisioning and risk analysis has been explored in a limited number of studies. Teixeira et al. [187] investigates the use of LLMs in generating credit risk reports, employing a few-shot prompting method. Their findings revealed that the LLM-generated reports were preferred over those produced by human credit analysts, demonstrating the efficacy of LLMs in this task. In a different approach, Cai [128] utilized FinGPT, an LLM fine-tuned on financial data, to generate synthetic data for enhancing supervised machine learning models in credit risk assessment. The analysis of financial documents by LLMs was explored in the work of Kim et al. [188], who showed the abilities of human analysts in predicting the earnings of business could be matched by GPT-4. Given the demonstrated capabilities of LLMs in processing and analyzing vast amounts of complex data, there is significant potential for LLM agents to enhance credit decisioning and risk analysis processes.

There has not been extensive research on credit pricing and scenario analysis using LLM agents, in the same way there has been with traditional agent-based modelling [8]. However, the previously mentioned work of Gao et al. [110] introduces an Agent-based Simulated Financial Market (ASFM) framework in a simulated stock market with an order-matching system mirroring real-world financial markets, simulating scenarios like interest rate cuts and inflation shocks, comparing the results with established economic theories to demonstrate the system's effectiveness.

3.5.4 Investment Banking Sales, Market Execution, and Mergers and Acquisitions

The application of LLM agents in investment banking sales, market execution, and market making has yet to receive substantial attention in academic publications. Nevertheless, several key tasks performed by investment banking salespeople are well-suited to benefit from LLMs and LLM agents. These tasks, as highlighted by financial institutions,[15] include researching investment news, analyzing market conditions, and synthesizing insights from earnings reports. Such activities, which require processing large volumes of textual data, align closely with the capabilities of LLM agents discussed in subsection 3.2. The limited work into mergers and acquisitions includes the work undertaken by Kropmans [189], which explored the use of LLM agents to recreate Information Memorandums (IMs). The IMs were evaluated and the authors concluded that "LLMs are very useful for internal and external analysis of companies". Future work in this domain could focus on how these roles can be automated by LLM agents to help ensure market availability and stability.

3.5.5 Commercial Banking and Institutional Advisory

The direct applications of LLM agents in commercial banking and institutional advisory remains relatively unexplored in current literature. Kikuchi et al. [190] investigates the capability of LLMs to analyse the outcomes of agent-based modeling simulations, finding LLMs can generate clear descriptions of simulation results and produce plausible business cases. Although this represents a limited application based on

[15] https://aws.amazon.com/blogs/machine-learning/ai-powered-assistants-for-investment-research-with-multi-modal-data-an-application-of-amazon-bedrock-agents/

simulated data, it demonstrates the potential of LLMs to derive actionable insights from complex datasets, a capability highly relevant to commercial banking. The research conducted by Tsao [191] introduces a sophisticated multi-LLM agent framework designed to assist corporations in conducting market landscape analyses, developing customer profiles, and formulating sales strategies. Utilizing a combination of historical and synthetic data, their findings demonstrate the efficacy of LLMs in enhancing the sales planning processes of businesses.

The ethical implications of using LLM agents in business contexts are highlighted by the work of Yu et al. [192], who found a decline in ethical decision-making when a LLaMA model was fine-tuned to prioritize economically beneficial outcomes. This finding underscores a significant concern in the potential deployment of LLM agents for business decisions, particularly in the sensitive domain of commercial banking. While current research in this area is limited, the demonstrated capabilities and identified challenges suggest that applications of LLM agents in commercial banking are likely to expand in the near future. As these technologies continue to evolve, it will be crucial to balance their potential benefits with careful consideration of ethical implications and regulatory requirements, as discussed in sec 4.2.

4 Challenges and Limitations

While the studies we reviewed in the previous section demonstrated the transformative potential of LLM agents in finance and banking, their integration into mission-critical systems is constrained by technical, regulatory, and ethical challenges. Some of these limitations are inherent in the underlying LLMs that are being used to power these agents, while others emerge from the unique challenges of applying LLM agents in finance and economics. These limitations threaten to undermine their reliability, fairness, and scalability in a sector where precision, compliance, and trust are paramount. In this section, we detail these limitations and the challenges in the deplyoment of LLM agents that the field faces.

4.1 Technical Challenges

Model selection. The selection of LLMs for agentic applications in finance and banking is fraught with trade-offs shaped by the vast choices of available models, the tension between general-purpose versatility and domain-specific efficiency of models, and the sector's stringent operational demands [14]. The breakneck pace of LLM development exacerbates this dilemma: frequent releases of existing models (e.g., OpenAI's o-series models), release of new models (e.g., DeepSeek), and obsoletion of fairly recent models (e.g., GPT-3.5) force institutions into cycles of validation and integration, in addition to requiring costly updates and reassessment of the LLM agent functions to align with updated capabilities and standards [15, 89]. This is especially true in domains such as LLM agent-based simulations [121, 123] and high-frequency trading [107, 110, 170], where institutions will need to constantly update to more powerful models to maintain competitive advantage. As the pace of release of more powerful models exponentially increases, financial institutions will need to balance adaptability, cost, and domain-specific performance while ensuring compliance with evolving regulatory standards. Another challenge listed above comes from the choice between general-purpose models like GPT-4o or LLaMa 3 and specialised models such as FinGPT [143] for their applications. While this choice is down to the target application, banks still need to conduct proper ethics and model validations to ensure relevance of the underlying model choice.

Context length limitations and memory. LLM agents typically operate within fixed context windows (e.g., GPT-4 Turbo: 128k tokens), limiting the amount of historical or transactional data they can process at one time. Even despite higher context windows available in newer models (e.g., GPT-4.1: 1M tokens), this is still a limitation for some applications [193]. For example, in finance and banking, where LLM agents may need to take actions based on longer contexts (e.g., multi-year 100-page SEC filings or years worth of longitudinal data), these limitations can impede the ability to capture the full context of a customer's financial behaviour or the understanding of a report, leading to suboptimal decision-making [194]. While architectures such as [108] have introduced memory modules (through short-term, medium-term, and long-term memory layers) for their LLM agents to manage and prioritise information efficiently, their efficacy across applications and context lengths still remains to be tested.

Integration with existing systems. Integrating LLM agents into the existing IT ecosystem of financial institutions is nontrivial. These institutions rely on complex, legacy systems that are not designed for the flexibility and scalability required by LLM agents but are deeply embedded within their operational frameworks [195]. This integration challenge is compounded by the need to maintain uninterrupted service during the transition period, where the non-availability of new AI-driven services could lead to operational disruption [16]. Seamless integration necessitates middleware solutions and API layers that can bridge modern AI models with traditional core banking systems.

4.2 Regulatory and Compliance

Adhering to the EU AI Act and other regulations. The EU AI Act classifies high-risk AI systems (e.g., credit scoring, fraud detection) as requiring stringent oversight, including transparency, human oversight, and robustness assessments [94]. In addition, other LLM agent applications described in the previous section, particularly those influencing customer outcomes (e.g., robo-advisory, loan underwriting), are likely to fall under these high-risk categories, necessitating rigorous documentation of model validation, behaviour, and decision logic [78]. There are numerous instances of banks receiving hefty fines for lack of transparency over automated decision-making of banking operations affecting customers (e.g., European Data Protection Board [196]), highlighting a key challenge in deploying LLM agent applications in the sector. This, in addition to regulations and directives by the U.S. SEC, the UK Financial Conduct Authority (FCA), and the Bank of England require the financial institutions to carefully navigate fragmented standards while still remaining innovative [95, 96, 197].

Copyright and intellectual property. The race to create the most powerful LLMs has led organisations to scrape and train LLMs on massive collections of loosely structured data without keeping a track of their original sources, creator intentions, or copyrighting or licensing status [198]. These dubious practices are creating an ethical, legal, and transparency crisis, with model developers facing multiple copyright lawsuits [199, 200]. To avoid this, while training language models or calibrating and validating their agents, financial institutions must ensure that the data does not violate copyright or intellectual property laws, and they must be able to trace and verify data sources. This requirement is critical in maintaining legal and ethical standards, especially when deploying models that make decisions impacting financial outcomes [201].

Privacy and data security. Handling sensitive financial data necessitates robust privacy and security measures. LLM agents, if not properly safeguarded, risk data breaches through inadvertent exposure of sensitive information. The challenge is twofold: securing the data during pre-training and fine-tuning of the model and ensuring that real-time interactions with the model do not compromise customer confidentiality or violate data protection laws [85]. An example of this is a chat agent, inadvertently sharing a different customer's data while interacting with a customer. However, LLMs are susceptible to adversarial attacks, such as prompt injections, which manipulate the inputs to the models to reveal sensitive data or execute malicious instructions. In banking, this can pose systemic risks, since for instance, a compromised LLM chatbot can leak sensitive customer information. While a number of systems incorporate guardrails to prevent this, LLMs are continue to be susceptible to attacks [85, 86]. LLM agent developers also need to be careful when using newer features being offered by the vendors such as "computer control", which enables LLMs to control the user's computer screen and web-browser (e.g., Anthropic's Claude 3.5 Sonnet and OpenAI's Operator Assistant). Such features are marred by issues like prompt-injection attacks from the LLM overriding user instructions with text on the screen, which is a huge security risk.

4.3 Ethics, Bias, and Fairness

Concentration of power. The development and deployment of LLMs in finance and banking are increasingly dominated by a handful of technology giants (e.g., OpenAI, Google, Meta) and well-funded startups (e.g., Anthropic, Deepseek). Even in the field of finance and banking, institutions like Bloomberg and S&P Global control exclusive financial datasets (e.g., real-time market feeds, earnings call transcripts), enabling them to build domain-specific models (e.g., BloombergGPT) that smaller competitors cannot replicate. This can lead to a series of challenges, such as banks facing unpredictable costs and operational risks to use these models to power their agents [202], larger firms influencing regulatory frameworks [78], and the users of these LLMs getting affected by the geopolitical tensions and data sovereignty issues that arise from the concentration of AI power in specific countries or regions [203].

Model herding. Another possible challenge for LLMs and LLM agents from dominance of a few players in the LLM space is that of model herding. In this context, model herding refers to more financial institutions using the same few LLMs, trained on similar data sources. This could lead to homogeneity in decision-making of LLM agents across institutions, which may result in diminishing strategies, for example, depletion of profitable market signals in financial trading and potentially reinforcing systemic biases across the industry due to reduced diversity in methodologies [204, 205].

Bias, fairness, and data leakage. Biases in training data (e.g., gender, racial, or socioeconomic biases) propagate into LLM outputs, while opaque decision-making undermines accountability. Biased loan approval recommendations or discriminatory credit scoring could violate fair lending laws (e.g., EU AI Act and U.S. Equal Credit Opportunity Act) [94]. Even apart from societal biases, the use of LLMs in simulations and trading (as detailed in the previous section), raises a significant challenge due to the risk of future lookahead bias during agent training and testing. A prominent example of this is the work of Park [76], where the authors use an LLM to identify the market anomalies of Black Monday in 1987, the financial crisis of 2008, and the shock at the start of the COVID-19 pandemic in 2020. These are three well known historical events, which means that they will have featured prominently in the training data of the LLM, meaning it is difficult to judge if the LLM really is effective at spotting anomalies. However, the limitation of data leakage is acknowledged by the work of Cheng and Chin [140], who limit their analysis to after the knowledge cutoff of their LLM.

4.4 Explainability and Transparency

The lack of transparency in third-party LLM providers complicates audits for bias, security, and alignment with financial regulations [90]. Moreover, LLMs' "black-box" nature conflicts with regulations like

GDPR's "right to explanation" and Basel III's model risk management requirements [83, 91]. Even with the opensource LLMs, the stochastic training processes and a lack of suitable methods to explain LLM outputs makes it very hard to explain LLM outputs [78]. There are numerous instances of banks receiving millions of euros worth of fines for lack of explainability over automated decisioning of banking operations affecting customers (e.g., European Data Protection Board [196]), highlighting a key challenge in deploying LLM agent applications in the sector.

4.5 Robustness and Reliability

Logic and numerical reasoning. LLMs struggle with precise arithmetic, probabilistic reasoning, and quantitative logic, often producing approximations or errors in arithmetic operations. Critical applications in finance and banking like portfolio optimisation, derivative pricing, and risk modelling demand exact numerical precision. Errors in cash flow calculations or mispriced options could lead to significant financial losses [81]. Hybrid architectures combining LLMs with symbolic calculators can mitigate this limitation but are not yet widely adopted.

Hallucination. A commonly observed drawback of LLMs is their tendency to output information that is false or misleading [206]. This leads to apprehensions regarding their use in applications such as automatic generation of financial reports that are subject to rigorous regulatory frameworks and statutory requirements, wherein inaccuracies or misrepresentations may precipitate severe repercussions for corporate entities. Similarly, widespread adoption of LLM agents in financial advisory roles may be subject to scrutiny since hallucinations may have an adverse impact on individuals' financial well-being. Several fact-checking tools such as GenAudit [207] have been developed for LLMs and LLM agents to mitigate hallucinations. Many agentic systems have also included sophisticated memory modules to include relevant information in their prompt. However, further research into hallucination and its impact on LLM agents across applications is an area of active research. Until LLM-generated content can be trusted to be error-free, researchers and users will need to monitor LLM-generated content and ensure its validity.

4.6 Lack of Benchmarks and Consistent Evaluation

Evaluation of LLM agents across applications. Evaluation of LLM agents in finance and economics remains fragmented. Different studies use a variety of tasks and metrics, so there is no single benchmark to compare how well an LLM's decisions align with human behaviour. For instance, Fan et al. [208] adapt canonical games (dictator game, Rock-Paper-Scissors, and ring-network game) to analyse LLMs, finding systematic deviations from human decision patterns . Likewise, Guo et al. [107] evaluate LLM preferences using competitive auction games, again revealing significant departures from human (economic) norms. These examples illustrate that different works rely on different game-based tests for "alignment", but no common measure exists. Without a shared benchmark or metric, results across studies cannot be directly compared or aggregated.

A further concern is that test data (news, reports, etc.) often overlap with an LLM's training set. For example, Lopez-Lira and Tang [209] explicitly restrict their evaluation to post-training news: their sample runs from Oct 2021 (after ChatGPT's September 2021 cutoff) to Dec 2022, ensuring that their evaluation is based on information not present in the model's training data . If evaluation uses headlines or media already seen by the model, metrics like prediction accuracy or trading returns will be misleadingly high. Robust evaluation thus requires "point-in-time" data splits or time-respecting LLM variants to avoid future lookahead bias.

The limitation in the evaluation of LLM agents for financial trading and is the nature of the metrics chosen for evaluation; typically only traditional metrics used in trading are used, like Sharpe Ratio and Maximal Drawdown. However, this removes the ability to evaluate on the intermediate steps that are undertaken by an agentic system, such as external data gathering and comparative analysis, and exclusively focuses on the final outcome, highlighted as a common limitation in the evaluation of many agentic systems by Zhuge et al. [210], warranting a new method for consistent benchmarks. A recurring issue among these works is the use of differing datasets for benchmarking performance and a lack of comparison to other LLM or LLM agent methods [130, 134], making it difficult to effectively compare the effectiveness of different approaches. There are exceptions to this, for example the work of Yu et al. [12] compares performance to the other LLM agent frameworks of Li et al. [144] and Zhang et al. [142] using the same benchmark datasets. Nie et al. [211] emphasise that most existing trading benchmarks and performance measures were designed before LLMs, and the "fundamental shift" requires new, LLM-tailored evaluation frameworks. In summary, relying only on traditional metrics (Sharpe, drawdown) may not fully assess an LLM agent's capabilities or alignment to human-like behaviour.

Calibration and validation of LLM agents. The field of classical agent-based modelling now contains a strong emphasis on developing models that are realistic. This presents itself both in the calibration of model parameters to ensure the simulation replicates historical values, as well as the validation of agent-based models. For example, the works of Carro et al. [212] and Hamill et al. [8] calibrate their models to historical values and undertake validation of their models using both stylised facts and time-wise validation. The calibration of the actions of LLM agents in simulation to replicate historical values does not seem to be present in the current literature. The validation of simulation fidelity with LLM agents has received some attention in the literature, though it often relies on qualitative assessments rather than rigorous quantitative metrics. For instance, Li et al. [115] evaluates their macroeconomic LLM agent simulation by qualitatively comparing trends in several macroeconomic indicators against classical agent modeling approaches

4.7 Alignment with Humans

Several of the research works presented in this paper aim to substitute humans with LLM agents for conducting large scale studies to assess the response of entities to a stimulus in the financial sector, for example, the response of agents to the launch of a new financial product. Works such as those of Gui and Toubia [87], Kitadai et al. [88], and Liu et al. [89] found that the responses of LLMs in financial scenarios are extremely rational and not human-like unless a lot of carefully-tailored context is provided. Hartley et al. [213] further enhance our understanding by investigating the relationship between LLMs' personality traits and risk propensity, finding that GPT-4o exhibits higher Conscientiousness and Agreeableness traits compared to human averages, while functioning as a risk-neutral rational agent in prospect selection. These finding highlights the shortcoming of the readiness of LLM agents in acting as autonomous agents and replacing humans in studies and simulations without human in the loop. Furthermore, findings in literature (e.g., Kitadai et al. [88]) that more powerful LLMs make choices more aligned with the optimal solution than the expected human decision, indicates that more powerful LLMs may not be the solution to this limitation, with sophisticated prompt engineering techniques being the sole mitigating strategy for this inherent limitation. Finally, studies such as Fan et al. [208] have highlighted that LLMs struggle with implicit reasoning, struggle with recognising the patterns in strategies of opponents, and even struggle with using the description of an opponent's optimal strategy unless it has been explicitly provided to them. These limitations raise questions on the readiness of LLMs to replace humans in key applications in the financial sector.

5 Future Prospects of LLM Agents in Banking and Finance

The integration of LLMs within finance and banking presents a landscape of substantial potential, poised to gain even greater significance in the immediate future. This section discusses the possible directions of progress for LLM agents in finance and banking, including technological advancements that can be integrated into agent systems, emerging industry applications, and the regulatory and ethical considerations that must accompany their development.

5.1 Emerging Research

Computer usage. A notable advance in the field of LLM agents is the demonstration of the abilities of LLMs to complete tasks using a computer, by observing what is currently on screen and interacting via a keyboard and mouse, as a human user would. Computer usage through LLMs is now offered by frontier labs, including Anthropic,[16] demonstrating the abilities of these LLMs to accomplish multi-step tasks. While these initial offerings are currently in beta, given the vast amount of modern work that happens via computer interaction, this could enable LLM agents to accomplish a much wider range of tasks.

Reasoning models. One of the most notable releases in LLM models recently have been reasoning models, LLMs which are specifically designed to undertake complex reasoning tasks. Notable releases include OpenAI's o1 and o3 models,[17] which have achieved notable scores in intelligence benchmarks, and DeepSeek's R1 model,[18] which has achieved similar levels of performance on benchmarks. These LLMs alone will likely be able to accomplish more complex tasks that previous generations of LLMs, through direct prompting or when powering single- or multi-agent frameworks.

One of the key considerations in the use of these reasoning models is the typically higher monetary cost and time taken for output compared to other LLMs. Rigorous evaluation of these LLMs on clear and comprehensive benchmarks will be required for financial professionals to determine whether the new abilities of these models are worth these costs.

Improvements in LLMs. Extensive research efforts have been dedicated to enhancing all aspects of the pipeline for developing and implementing large language models. Alternative action-reward methods for LLM alignment are being researched, including credit assignments for tokens [214], distributional presence alignment [215], and inverse reinforcement learning [216]. These could potentially align LLM agent decision-making closer to human preferences. Exploring established and novel prompting methods like chain-of-thought [155], tree-of-thought [217], Multi-Agent System for conditional Mining (MACM) [218], and Chain of Mathematically Annotated Thought (CoMAT) [219] could enhance LLM agents' performance in finance and economics, particularly in mathematical and financial reasoning tasks. Research into agent interactions, including new communication paradigms [220], task decomposition methods [221], and world model integration [222], has shown potential to enhance LLM agent performance, potentially extending into finance and banking.

Adaptive and continuous learning in LLM agents. Exploration of LLM agents that are adaptive and can learn continuously over time is an active area of research. The self-evolving abilities of LLM agents are

[16] https://docs.anthropic.com/en/docs/build-with-claude/computer-use
[17] https://openai.com/index/openai-o3-mini/
[18] https://api-docs.deepseek.com/news/news250120

presented in the work of Guan et al. [223] and through automatically tuned prompts of active agents in the work of Bo et al. [224]. These abilities are of note for finance and banking as LLM agents will be exposed to a lot of data in their environments, so they have a substantial opportunity to continuously improve without human intervention.

Improved benchmarks and performance measurement. Sustained progress in applications of LLM agents to financial and economic applications will be greatly helped by the consistent use of rich benchmarking datasets in the field. Several finance and economic benchmarks have emerged, like PIXIU [225] and FinBen [226], but have not been widely adopted in LLM agent studies. Additionally, benchmarks for evaluating LLMs' ability to roleplay [227], could enhance the assessment of LLM capabilities in financial contexts.

Developing financial decision-making benchmarks and multi-modal evaluations [12, 228] will advance LLM agents in finance. Granular performance assessments, like in the DevAI benchmark [210], will provide deeper insights into agent capabilities.

5.2 Deployment and Integration within Banking and Finance

The integration of LLM agents in finance and banking presents practical challenges requiring further investigation. Research is needed to determine customer preferences for LLM-assisted services and to optimise LLM performance for time-sensitive applications like high-frequency trading. Robust deployment practices ensuring data confidentiality are essential, potentially leveraging secure multi-party computation and privacy-preserving AI techniques.

A comprehensive cost-benefit analysis of LLM agent deployment is crucial, considering implementation costs and potential value added through improved efficiency and accuracy. Developing standardised metrics for assessing return on investment would aid decision-makers in determining optimal deployment strategies. Addressing these considerations through rigorous research and industry collaboration is vital for the successful and responsible integration of LLM agents in finance and banking.

5.3 Developing Ethical and Trustworthy Agents

Future research should focus on developing LLM agents that align with human ethics, resist jailbreaking, and mitigate risks to financial system stability. Ensuring equitable outcomes from LLM agent decisions requires dedicated research to identify and mitigate biases, while improving transparency of decision-making processes through interpretable AI models and user-friendly interfaces. Establishing a comprehensive set of ethical financial regulations tailored for LLM agents is necessary, addressing issues such as accountability, data privacy, and the appropriate scope of LLM agent autonomy in financial decision-making.

The future of LLM agents in finance and economics requires addressing key challenges: developing comprehensive benchmarks, improving alignment with human decision-making, enhancing safety measures, exploring multi-modal capabilities, addressing data leakage, and improving cost-effectiveness. By focusing on these areas, researchers can create more reliable, transparent, and ethically sound LLM agents that complement human expertise in financial and economic domains.

6 Conclusion

This survey provides a comprehensive overview of current LLM-agent research in the financial sector, highlighting how these agents serve as simulators, analysts, advisors, and actors.

In their role as simulators, LLM agents have been employed in auction and trading simulations, macro- and micro-economic modeling, and synthetic data generation. As analysts, they have been applied to market and sentiment analysis, stock-price prediction, document review, and compliance monitoring. In their capacity as advisors, they support automated customer service, financial planning and budgeting, and decentralised-finance applications. Finally, as actors, they execute tasks such as algorithmic trading, portfolio management, asset pricing, and direct customer interaction. We also identify several promising but underexplored applications of LLM agents.

Despite these advances, LLM agents inherit the limitations of their underlying language models and face integration challenges within legacy financial-institution systems. Moreover, the high-stakes nature of finance raises critical concerns around ethics, regulation, and fairness. As LLMs grow more capable and agent architectures become more sophisticated, we hope this review will spur further research into both the potential and the risks of deploying agentic LLMs, ultimately enabling humans to assume supervisory roles over increasingly autonomous, personalised, and scalable financial services.

Acknowledgments

We wish to express appreciation to Graham Smith and Zachery Anderson of NatWest Group for the time and support needed to develop this research paper.

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